WILSON S. MARTINS
Curriculum Vitae

PERSONAL INFORMATION
Orcid: 0000-0002-8425-1311
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CURRENT POSITION: Postdoctoral research assistant, UMass Boston.

EDUCATION

05/2022 – 02/2026 **Ph.D. in Physics**, University of Tübingen, Germany.
Thesis: *Driven systems based on trapped Rydberg atoms and ions: applications and thermodynamics.* Graduated with honors *(magna cum laude)*.
Advisor: Prof. Dr. Igor Lesanovsky

02/2020 – 03/2022 **M.Sc. in Physics**, University of São Paulo, Brazil.
Thesis: *Suppressing information storage in a structured thermal bath*.
Grade: A/A
Advisor: Prof. Dr. Diogo O. Soares-Pinto

03/2016 – 11/2019 **B.Sc. in Physics**, University of São Paulo, Brazil.
Thesis: *Study of an effective spin model using Monte Carlo Simulations*.
Advisor: Prof. Dr. Eric C. Andrade

PUBLICATIONS (Scholar)

[1] *Impact of micromotion and field-axis misalignment on the excitation of Rydberg states of ions in a Paul trap.*
Wilson S. Martins, Joseph Wilkinson, Markus Hennrich, and Igor Lesanovsky.
Phys. Rev. A **111**, 043106 – 04/2025 (link) (***Editors' Suggestion***)

[2] *Quasiperiodic Floquet-Gibbs states in Rydberg atomic systems.*
Wilson S. Martins, Federico Carollo, Kay Brandner, and Igor Lesanovsky.
Phys. Rev. A **111**, L010202 – 01/2025 (link)

[3] *Rydberg-ion flywheel for quantum work storage*.
Wilson S. Martins, Federico Carollo, Weibin Li, Kay Brandner, and Igor Lesanovsky.
Phys. Rev. A **108**, L050201 – 11/2023 (link)

[4] *Quantum simulation with Rydberg ions in a Penning trap. (pre-print)*
Wilson S. Martins, Markus Hennrich, Ferdinand Schmidt-Kaller, and Igor Lesanovsky.
arXiv:2601.01626 – 01/2026 (link)

SELECTED TALKS & PRESENTATIONS

2024 – Bridging Physics & Mathematics of Quantum Chaos, Helsinki — **Talk:** *Quasiperiodic Floquet-Gibbs states in Rydberg atomic systems.*

2023 – 13th Nottingham Symposium on Quantum Systems — **Invited talk:** *A Rydberg ion quantum engine*.

2023 – DPG SAMOP Conference, Hannover — **Talk:** *A Rydberg ion quantum engine*.

2024 – Early Career Conference in Trapped Ions (ECCTI) — **Poster:** *Floquet-Gibbs states in laser-driven atomic systems*.

TEACHING EXPERIENCE & PUBLIC OUTREACH

• Interview to New Science Magazine about publication [3]. *Quantum flywheel could be fashioned from super-sized charged atoms.*

• Tutor – *Classical Field Theory*, University of Tübingen (2023-2024)

• Tutor – *Condensed Matter Quantum Many-Body Field Theory*, University of Tübingen (2023–2024)

• Tutor – *Statistical Physics and Thermodynamics*, University of Tübingen (2022–2023)

• Tutor – *Physics Laboratory I & II*, University of São Paulo (2018–2021)

REFERENCE CONTACTS

Prof. Dr. Igor Lesanovsky (igor.lesanovsky@uni-tuebingen.de) – Ph.D. Supervisor
Prof. Dr. Diogo O. Soares-Pinto (dosp@ifsc.usp.br) – M.Sc. Supervisor
Prof. Dr. Kay Brandner (kay.brandner@nottingham.ac.uk) – Collaborator